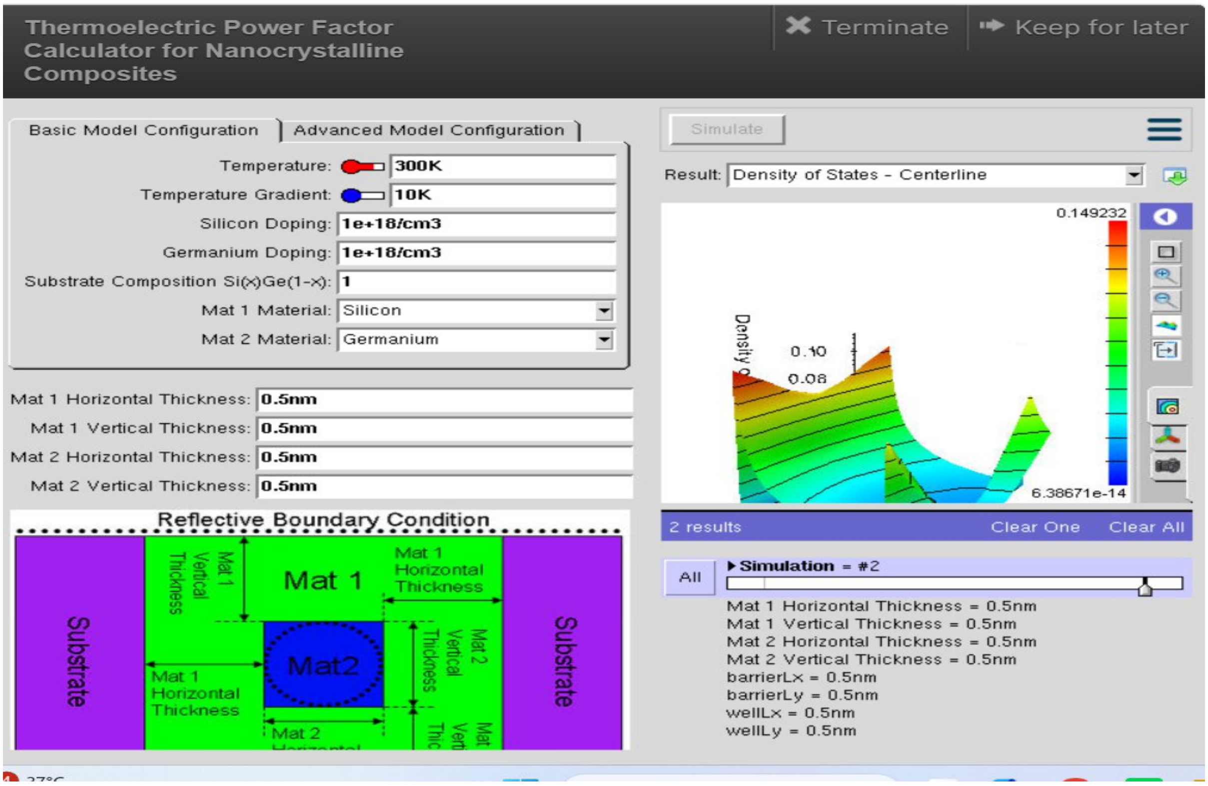
Exp. 2. Simulation of Density of States of Quantum Well Structure with Various Nanometer Thickness.

Aim: To simulate density of states of quantum well structure with various nano meter thickness

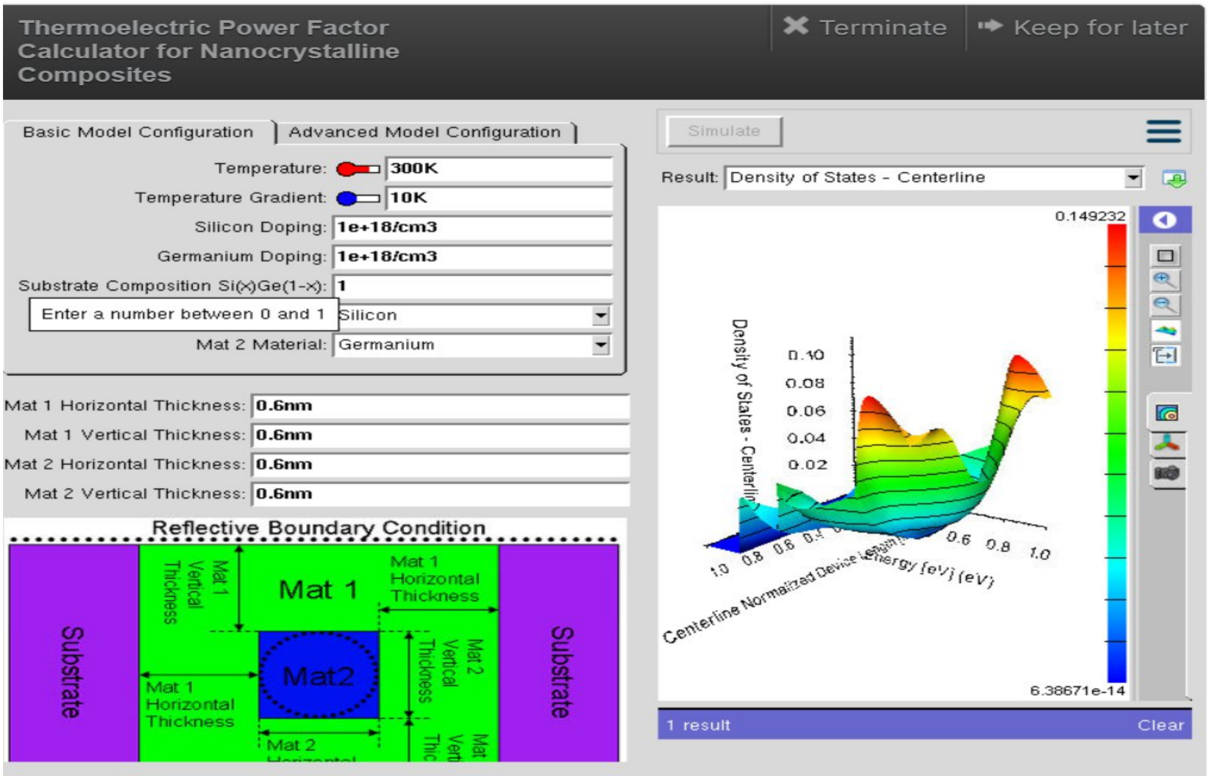
Case Study 1: Simulate the density of states of quantum well of thickness 0.5 nm



Observation:

- The quantum well thickness was set to 0.5 nm while keeping all other parameters constant (Temperature = 300 K, Temperature Gradient = 10 K, Silicon and Germanium doping = 1×10^{18} cm^{-3}).
- The simulated Density of States (DOS) plot shows sharp and distinct peaks, indicating strong quantum confinement.
- The energy levels are widely separated due to the very thin quantum well.
- Charge carriers are confined to a smaller region, resulting in discrete energy states rather than continuous bands.
- The DOS distribution changes rapidly with energy, indicating significant quantization effects.

Case Study 2: Simulate the density of states of quantum well of thickness 0.6 nm



Observation:

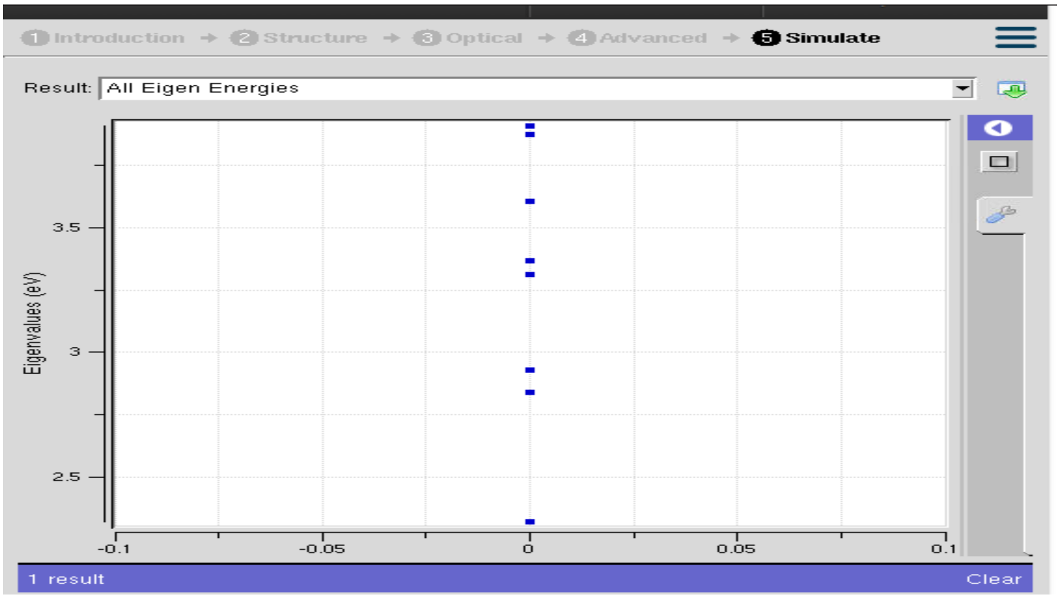
- The quantum well thickness was increased to **0.6 nm**, while all other simulation parameters remained unchanged.
- The DOS graph shows **broader and smoother peaks** compared to the 0.5 nm case.
- The spacing between quantized energy levels decreases because the wider quantum well reduces quantum confinement.
- More energy states become available for carrier occupation, leading to a smoother DOS profile.
- Carrier confinement is slightly weaker than in the 0.5 nm structure.

Conclusion

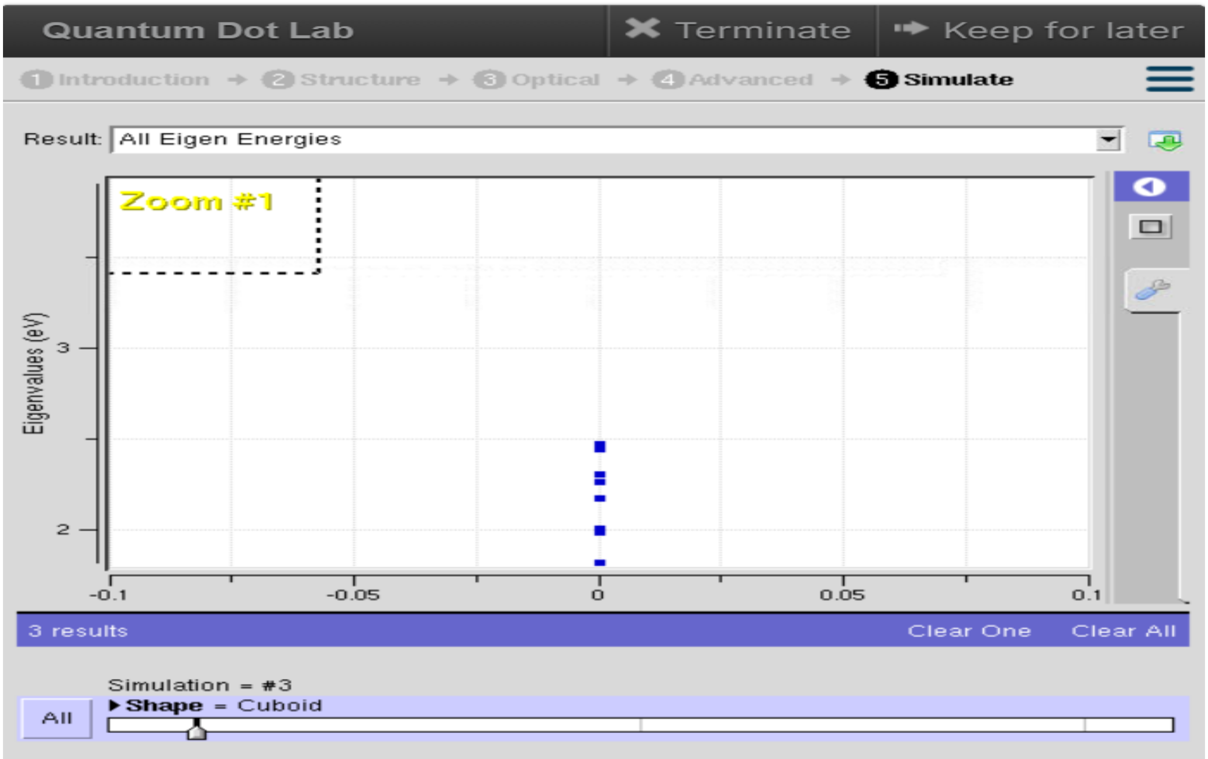
The simulation demonstrates that quantum well thickness has a significant influence on the Density of States (DOS). At a thickness of 0.5 nm, strong quantum confinement produces well-separated discrete energy levels and sharp DOS peaks. Increasing the thickness to 0.6 nm reduces the confinement effect, resulting in closer energy levels and smoother DOS characteristics. Thus, as the quantum well becomes thicker, the electronic behavior gradually approaches that of a bulk semiconductor.

Exp. 3. Simulation of Energy States of Quantum Dot using NanoHub

Aim: To obtain the energy states of Quantum dots.



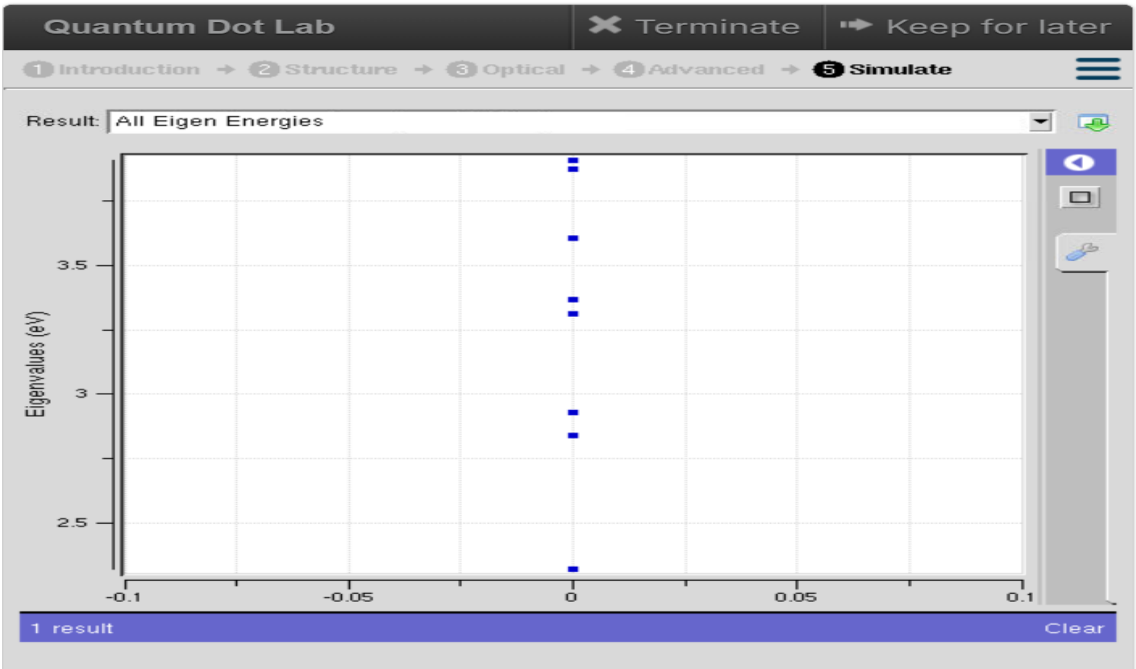
Case Study 1: Simulate the energy states of cylindrical quantum dots of X=15 nm, Y = 10.5nm and Z = 8nm.



Observation

- The NanoHub simulation displays several discrete eigen energy levels, confirming quantum confinement inside the cylindrical quantum dot.
- The energy states are distributed approximately between 2.3 eV and 3.8 eV.
- Lower energy levels are closely spaced, whereas higher energy levels show comparatively larger separation.
- No continuous energy band is observed, indicating the presence of quantized energy states.
- The cylindrical geometry confines charge carriers in all three dimensions, resulting in well-defined allowed energy levels.
- The finite number of eigenstates indicates that only specific electron energies are permitted within the quantum dot.

Case Study 2: Simulate the energy states of two layer cuboid quantum dots



Observation

- The simulation produces multiple **discrete eigen energy levels** , characteristic of quantum confinement.
- The energy levels are observed approximately between **1.8 eV and 2.5 eV** , which are lower than those of the cylindrical quantum dot.
- The spacing between adjacent energy levels is relatively small, indicating stronger interaction between the two quantum dot layers.
- The two-layer cuboid structure introduces additional confinement effects that modify the electronic energy spectrum.
- The energy levels remain discrete without forming continuous bands, confirming nanoscale confinement.

Conclusion

- The NanoHub simulation successfully demonstrates the formation of **quantized energy states** in quantum dots due to three-dimensional quantum confinement.
- Both the cylindrical and two-layer cuboid quantum dots exhibit discrete eigen energies rather than continuous energy bands.
- The cylindrical quantum dot shows higher-energy eigenstates with comparatively larger spacing between levels, while the two-layer cuboid quantum dot exhibits lower energy states with closer spacing due to the influence of the layered structure.
- These results confirm that the **shape, dimensions, and structural configuration of a quantum dot significantly influence its electronic energy spectrum** , making quantum dots highly suitable for applications

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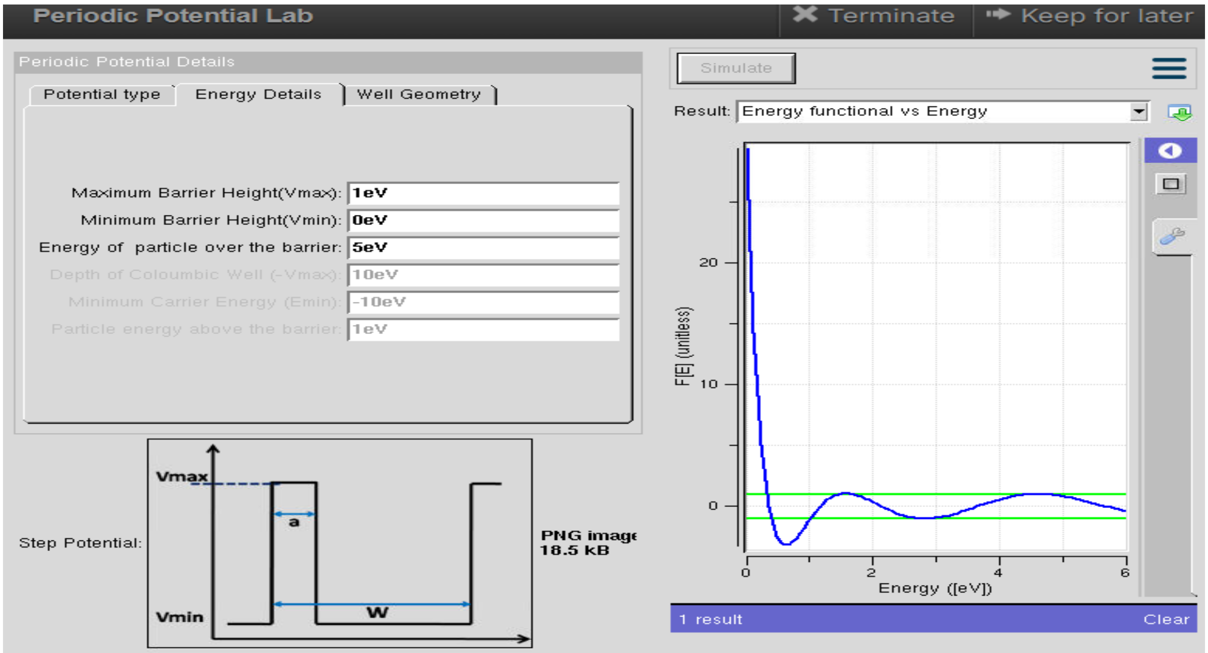
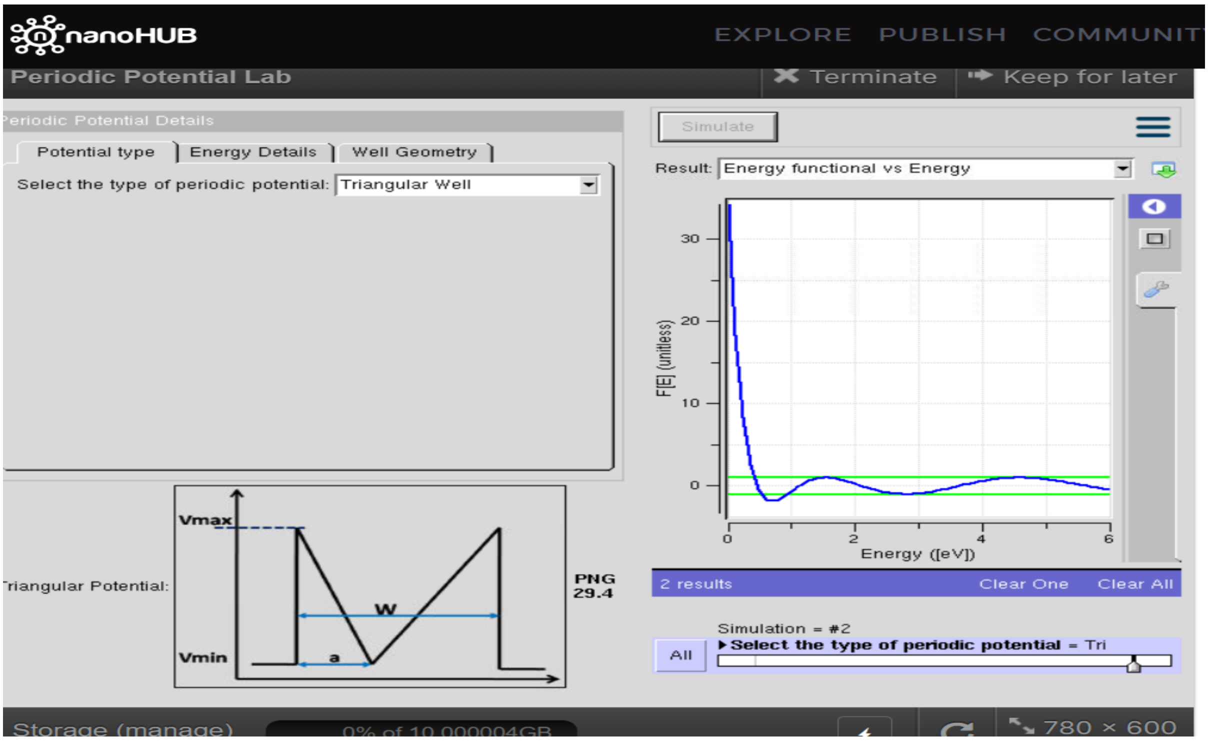
Exp. 4. Kronig Penney Model

Aim: To simulate Energy functional vs Energy graph using Kronig Penney model



Case Study 1: Simulate the output for Triangular well



Case Study 2: Simulate the output for Sinusoidal Potential